

Week 1

Introduction to Database

Chapter 1 - Part 1

Data: known facts that can be recorded and have an implicit meaning.

Database: A collection of related data. It has the following implicit properties:

A database represents some aspects of the real world, sometimes called the miniworld.

A database is a logically coherent collection of data with some inherent meaning.

A database is designed, built and populated with data for specific purpose.

Ex: Airline reservation system, Students' registration system.

Database Management System (DBMS):

A software package/system to facilitate the creation and maintenance of a computerized database.

Database System: The DBMS software together with the data itself. Sometimes the applications are also included.

There are two approaches to manage data :
(traditional)

File - based approach - An approach that utilizes a collection of application programs which performs services to end - users (eg. Reports). Each program defines and manages its own data. — Ex: wordpad, notepad, etc.

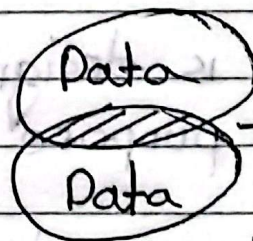
(new)
Database Approach - An approach that data is collected and manipulated using specific software called Database Management System, and many programs share this data.

Drawbacks :

File Based approach

User 1 Application programs

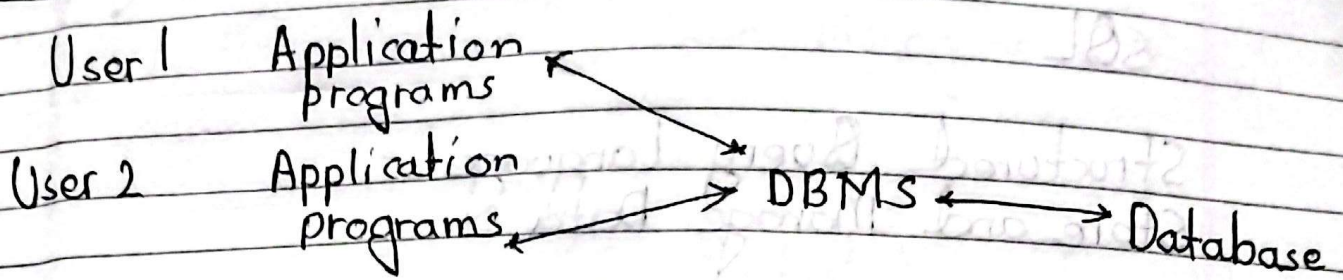
User 2 Application programs



— common data repeated in multiple data files

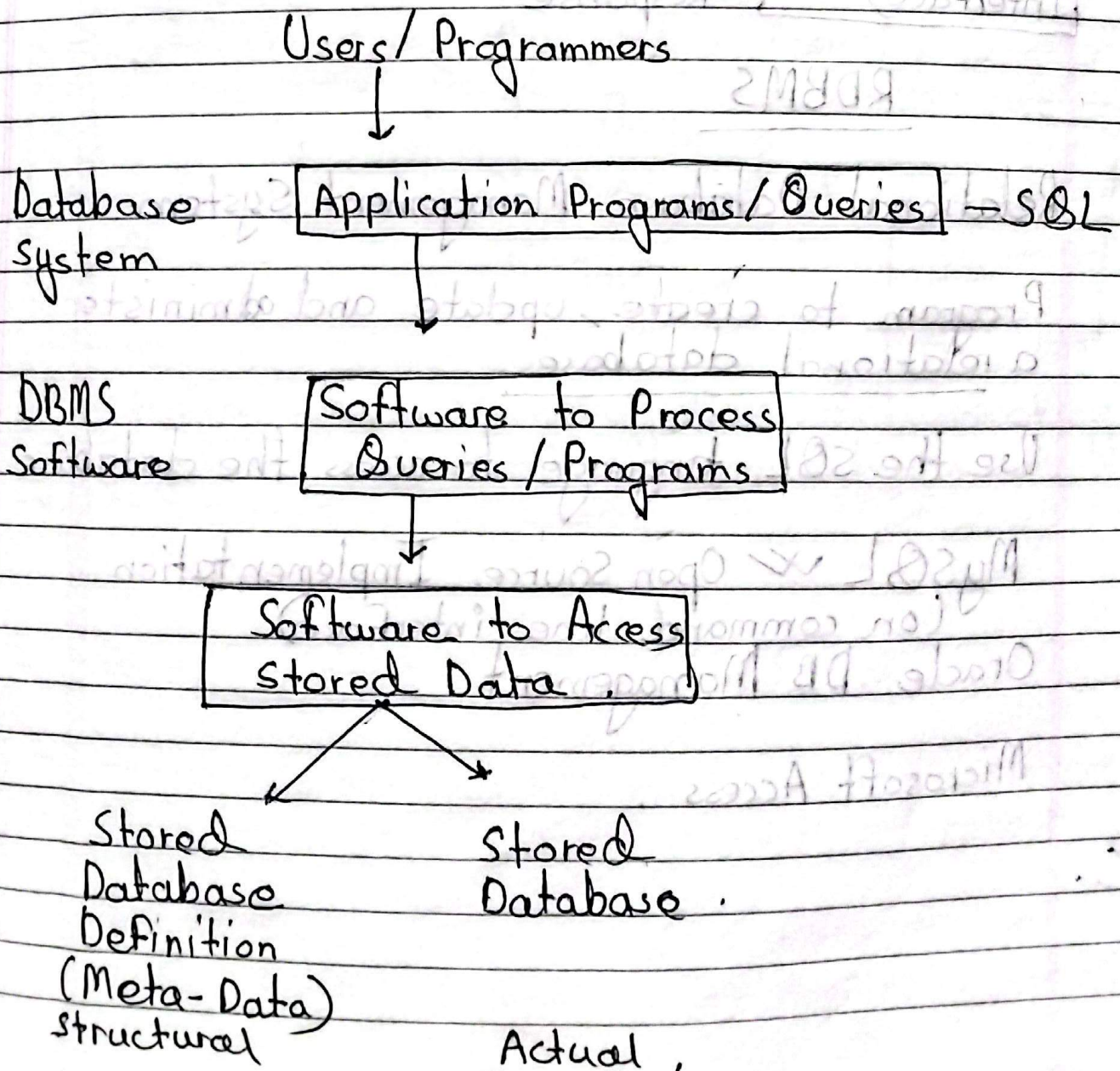
Waste of storage space
Data maintenance

Database Approach



All data is stored only once no wastage of space.
One standard Format.

Simplified database system environment



Functionality and Characteristics of Database:

Typical DBMS Functionality

Define a particular database in terms of its data type, structures and constraints.

Construct or load the initial database contents on a secondary storage medium.

Manipulating the database:

Retrieval: Querying, generating reports.

Modification: Insertion, deletion and updates to its contents.

Accessing the database through web applications.

Processing and Sharing by a set of concurrent users many users connecting and using database at same time and application programs - yet keeping all data valid and consistent.

Other features:

Protection or security measures to prevent unauthorised access.

Maintaining the database and associated programs over the lifetime of database application.

Main Characteristics of the Database Approach

Self-describing nature of a database system:

A DBMS catalog stores the description of a particular database (eg. data structures, types and constraints)

The description is called meta-data.

This allows the DBMS software to work with different database applications.

Insulation between programs and data:

Called program-data independence.

Allows changing data structures and storage organization without having to change the DBMS access programs.

Support of multiple views of the data:

Each user may see a different view of the database; which describes only the data of interest to that user.

Database Users and Use of Database

Database Users :

Actors on the Scene : They actually use and control the database content, and design, develop and maintain database applications.

Database Administrators

Database Designers

Software Engineers

End - Users

Workers Behind the Scene :

Those who design and develop the DBMS software and related tools.

Database Administrators

are responsible for authorizing access to the database, for coordinating and monitoring its use, acquiring software and hardware resources, controlling its use and monitoring efficiency of operations.

Database Designers

Responsible to define the content, the structure, the constraints and functions or transactions against the database. They must communicate with the end-users and understand their needs.

Categories of End-Users :

Actors on the Scene :

End-Users : Are those who require access to the database for querying, updating and generating reports. They are categorized as :

Casual end-users : occasionally access the database

Naïve or parametric : constantly update and query databases, using standard types of queries and updates

Sophisticated end-users : thoroughly familiarize themselves with facilities of DBMS so as to implement their application to meet their complex requirements

Stand alone end-users : maintain personal databases by using easy to use ready made program packages

Advantages of Using Database Approach :

Controlling duplication of data without need redundancy in data storage and in development and maintenance efforts.

Sharing of data among multiple users.

Restricting unauthorized access to data.

Providing persistent storage for program objects.

In Object - Oriented DBMS.

Week-3 Data Models

Database System Concepts and Architecture

Data Abstraction:

The suppression of details of data organization and storage and the highlighting of essential features for better understanding.

Data Model:

A set of concepts to describe the structure of a database, and certain constraints that the database should obey.

Data model helps to achieve data abstraction.

Categories of Data Models

Conceptual (high-level, semantic) data models:

Provide concepts that are close to the way many users perceive data.

Also called entity-based or object-based data models.

Physical (low-level, internal) data models:

Provide concepts that describe details of how data is stored in the computer.

Representational (Implementation) data models:

Provide concepts that may be understood by end users but that are not too different from the way data is organized within the computer.

Hide some details of data storage but can be implemented on a computer system in a direct way.

Used by many commercial DBMS implementations (eg. relational data models used in many commercial systems).

Schemas versus Instances

Database Schema:

The description of a database.

Includes description of the database structure, data types and the constraints on the database.

Schema Diagram:

An illustrative display of a database schema.

Schema Construct:

A component of the schema or an object within eg the schema eg. STUDENT, COURSE.

Database state:

The actual data stored in a database at a particular moment in time. This includes the collection of all the data in the database.

Also called database instance (or occurrence or snapshot).

Example of a database schema:

STUDENT - All caps

Name	Student number	Class	Major
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Database Schema Vs Database State

Database State:

Refers to the content of a database at a moment in time.

Initial Database State:

Refers to the database state when it is initially loaded into the system.

Valid State:

A state that satisfies the structure and constraints of the database.

Schema and State Distinction:

The database schema changes very infrequently. The database state changes everytime the database is updated.

~~St~~ Schema is also called intension whereas State is also called extension.

Three Schema Architecture :

Advantage of using three schema architecture :

If change is made in one level suppose the no. of bits for student ID changed from 8 bits to 16 bits in internal schema, it will not affect conceptual schema. - Physical Data Independence.

DBMS :

Data Definition Language :

Ex : create database ;
create tables ;

DML : insert, delete, modify, select

Concurrency control : DBMS can support many users at the same time concurrently.

Three Schema Architecture of DBMS:

Schema translates to Structure

We need this three Schema Architecture for "Data Independence".

User and data are related but they should not interact directly.

User shouldn't know where the data is stored / how it is stored.

- Data Abstraction.

Example: Gmail

Date:

External Schema → View level
(multiple)



Conceptual Schema → Logical level



Internal Schema



Database (Disk)

External Schema - View level is given to the users but "all users will not have the same view".

Example: Connect (Student view
faculty view)

(Blueprint)

Conceptual Schema - How we are representing the data (describe the structure)

Example : Student (ID, c_gpa, dept)

Later on you will see ER Model, EER Model, Schema

Relationship between the Tables

Student & Course Table

Also the constraints for database will be mentioned.

Internal Schema - which files we have used to store the data/ what are the sizes of the files. Where the data is physically present → access paths (indexes)

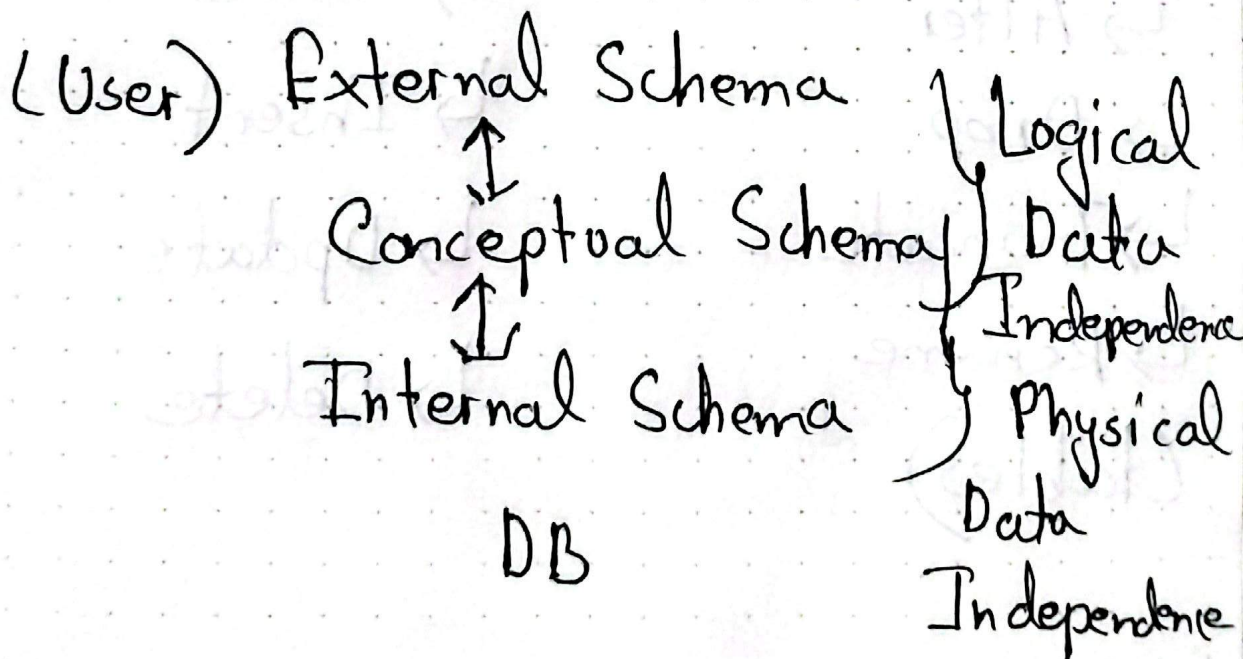
Date:

Frontend → external schema

Database Designer → conceptual schema
(blueprint)

Database Administrator → internal
(They have the whole access) Schema

Data Independence



SQL Commands

Data Definition

Language

(DDL)

Deals with
schema

↳ Create

↳ Alter

↳ Drop

↳ Truncate

↳ Rename

(Tables)

Data Manipulation

Language

(DML)

Deals with
changes in
data

↳ Select

↳ Insert

↳ Update

↳ Delete